Guided Problem Solving – 4

Full Worked Solutions

Foundations of Fluid Mechanics, AM2530
(B.Tech level – linear/angular momentum and dimensional analysis)

September 15, 2026

Problem 1 — Torque to Hold a Lawn Sprinkler Stationary

Given: Gage pressure = 140 kPa, total flow rate $Q = 4$ L/min, each jet exit speed *relative to the sprinkler arm* $V_{rel} = 17$ m/s, jet inclined 30° above horizontal, arm radius $R = 200$ mm, bearing friction torque $T_f = 0.18$ N · m (opposing rotation). Find the torque T_{hold} needed to keep the sprinkler stationary ($\omega = 0$).

Concept Refresher

For a control volume enclosing the sprinkler head, the **angular-momentum theorem** about the (vertical) rotation axis states

$$\sum T_{ext} = \dot{m}\,(rV_t)_{out} - \dot{m}\,(rV_t)_{in}$$

where V_t is the *absolute* tangential velocity component of the flow at each section, and r is the radial distance from the axis. The inlet flow travels along the vertical axis itself, so it carries zero angular momentum in ($r_{in} = 0$).

Step 1: Convert the flow rate

$$Q = 4 \text{ L/min} = \frac{4 \times 10^{-3} \text{ m}^3}{60 \text{ s}} = 6.667 \times 10^{-5} \text{ m}^3/\text{s}$$

$$\dot{m} = \rho Q = (1000)(6.667 \times 10^{-5}) = 0.06667 \text{ kg/s}$$

Step 2: Consistency check on the jet speed (optional, using the given pressure)

Applying Bernoulli's equation across the nozzle (stagnation gage pressure 140 kPa converting entirely to jet kinetic energy, elevation change negligible):

$$V_{jet} = \sqrt{\frac{2\,p_{gage}}{\rho}} = \sqrt{\frac{2(140{,}000)}{1000}} = \sqrt{280} = 16.7 \text{ m/s}$$

This is very close to the given $V_{rel} = 17$ m/s, confirming the data are self-consistent (the small difference is due to minor losses in the nozzle). The pressure value is therefore *not* separately needed in the torque calculation below – it is supplementary information.

Step 3: Absolute tangential velocity of the exiting jets

Since we want the torque to hold the sprinkler **stationary** ($\omega = 0$), the arm itself is not moving, so the *absolute* velocity of the water leaving each nozzle is identical to the velocity *relative to the arm*: $V_{abs} = V_{rel} = 17$ m/s, inclined 30° above horizontal.

Only the **horizontal** component of this velocity is tangential to the circle of rotation (perpendicular to the arm, in the horizontal plane) and hence contributes to torque about the vertical axis; the vertical component of the jet has no moment arm about a vertical axis passing through the arm.

$$V_t = V_{rel} \cos 30^\circ = 17 \times 0.8660 = 14.72 \text{ m/s}$$

Step 4: Torque exerted by the jet reaction (angular momentum theorem)

Summing the contribution of *both* jets is equivalent to using the total mass flow rate $\dot{m}$ with the common radius R and common tangential speed V_t (by symmetry, each arm carries $\dot{m}/2$, and $2 \times (\dot{m}/2)RV_t = \dot{m}RV_t$):

$$T_{jet} = \dot{m}\, R\, V_t = (0.06667)(0.2)(14.72)$$

$$T_{jet} = 0.1963 \text{ N} \cdot \text{m}$$

This is the torque that the escaping water reaction would apply to *spin* the sprinkler if it were left free (this is exactly the driving torque of an operating sprinkler).

Step 5: Balance of torques to keep the sprinkler stationary

To prevent rotation, an externally applied holding torque T_{hold} must balance the jet-reaction torque. The bearing-friction torque T_f *also* opposes the impending rotation (it always resists relative sliding, and here the impending motion is in the same sense as the jet reaction), so it assists the holding torque:

$$T_{hold} + T_f = T_{jet}$$

$$T_{hold} = T_{jet} - T_f = 0.1963 - 0.18$$

Final Answer

$$\boxed{T_{hold} \approx 0.0163 \text{ N} \cdot \text{m}}$$

Only a small applied torque is needed, because bearing friction alone supplies most of the resistance required to counter the jet reaction.

Problem 2 — Jet Striking a Moving Two-Armed Vane on a Cart

Given: A free jet (ρ, V_j, A_j) strikes a vane mounted on a frictionless cart moving at constant velocity V_c (to the right, same direction as the jet). The vane symmetrically splits the flow into two equal streams, each leaving at angle θ (above and below the horizontal, relative to the vane/ cart) so that by symmetry the net vertical force $F_y = 0$. Find (a) F_x (the restraining force), (b) the power delivered to the cart, and the cart speeds that (c) maximize F_x and (d) maximize the power.

Concept Refresher

Because the cart moves at **constant** velocity V_c, a reference frame attached to the cart is still an **inertial frame** (no acceleration), so Newton's second law / the linear-momentum theorem can be applied directly in this moving frame using *relative* velocities.

Step 1: Relative velocity seen by the vane

In the frame moving with the cart, the jet approaches the vane at the relative speed

$$V_{rel} = V_j - V_c$$

Only fluid that is actually catching up to the (already-moving) cart interacts with the vane, so the mass flow rate processed by the vane (in the moving frame) is

$$\dot{m}_{rel} = \rho A_j \, (V_j - V_c)$$

Step 2: Frictionless vane – relative speed is preserved

Since the vane is frictionless and we neglect elevation change and losses, Bernoulli's equation in the moving (inertial) frame shows the fluid leaves the vane at the *same relative speed* it arrived with, only turned through angle θ: exit relative speed $= V_j - V_c$ for each of the two branches, each directed at $\pm\theta$ from the horizontal.

Step 3: x-momentum balance to find F_x

The x-component of the exit relative velocity for *either* branch is $(V_j - V_c)\cos\theta$ (the same for both, since $\cos(+\theta) = \cos(-\theta)$); the y-components are equal and opposite and cancel when summed, consistent with $F_y = 0$.

$$F_x = \dot{m}_{rel}\big[(V_j - V_c) - (V_j - V_c)\cos\theta\big] = \dot{m}_{rel}(V_j - V_c)(1 - \cos\theta)$$

$$\boxed{F_x = \rho A_j (V_j - V_c)^2 (1 - \cos\theta)}$$

This is the force that must be applied to the cart (by whatever restrains it) to keep it moving at the constant velocity V_c; by Newton's third law it also equals the thrust that the jet exerts on the cart.

Step 4: Power delivered to the cart

Power = force × velocity of the point of application (the cart moves at V_c):

$$\boxed{P = F_x V_c = \rho A_j (V_j - V_c)^2 (1 - \cos\theta)\, V_c}$$

Step 5(c): Cart velocity that maximizes F_x

$$F_x(V_c) = \rho A_j(1-\cos\theta)(V_j - V_c)^2$$

This is a strictly decreasing function of V_c for $0 \le V_c < V_j$ (since $(V_j - V_c)^2$ is largest when V_c is smallest). Hence

Final Answer

$$F_x \text{ is maximum at } V_c = 0 \qquad \Longrightarrow \qquad F_{x,max} = \rho A_j V_j^2(1-\cos\theta)$$

(the physically obvious case: the force on a *stationary* vane is the largest possible, since the cart extracts no momentum "credit" by moving away from the jet.)

Step 6(d): Cart velocity that maximizes the power

$$P(V_c) = \rho A_j(1-\cos\theta)\, V_c(V_j - V_c)^2$$

Differentiate with respect to V_c and set to zero:

$$\frac{dP}{dV_c} = \rho A_j(1-\cos\theta)\Big[(V_j - V_c)^2 + V_c \cdot 2(V_j - V_c)(-1)\Big]$$

$$= \rho A_j(1-\cos\theta)(V_j - V_c)\Big[(V_j - V_c) - 2V_c\Big] = \rho A_j(1-\cos\theta)(V_j - V_c)(V_j - 3V_c)$$

Setting this to zero gives two roots: $V_c = V_j$ (trivial – the cart moves as fast as the jet, so no relative flow at all, and $P = 0$, a minimum) and

$$V_j - 3V_c = 0 \quad \Longrightarrow \quad V_c = \frac{V_j}{3}$$

A quick check of the sign of dP/dV_c on either side of $V_c = V_j/3$ confirms this is a **maximum**.

Final Answer

$$\boxed{V_c = \frac{V_j}{3}}$$

Maximum power, obtained by substituting back:

$$P_{max} = \rho A_j(1-\cos\theta)\Big(\frac{V_j}{3}\Big)\Big(V_j - \frac{V_j}{3}\Big)^2 = \rho A_j(1-\cos\theta)\Big(\frac{V_j}{3}\Big)\Big(\frac{2V_j}{3}\Big)^2$$

$$\boxed{P_{max} = \frac{4}{27}\rho A_j V_j^3(1-\cos\theta)}$$

Physical Check

The result $V_c = V_j/3$ for maximum power is the classical result for jet-propulsion/turbine-blade problems of this type (it is the same principle used, e.g., in Pelton-wheel bucket analysis), and matches the well-known textbook answer for this exact configuration.

Problem 3 — Weir Flow-Rate Formula by the Pi Theorem

Given: $Q = f(g, b, H)$, where Q is volume flow rate over a weir, g is gravitational acceleration, b is the weir width (into the page), and H is the upstream water height above the weir crest. It is additionally known that $Q \propto b$. Find the functional relationship $Q(g, b, H)$.

Step 1: List variables and dimensions

Variable	Symbol	Dimensions (MLT)
Flow rate	Q	L^3T^{-1}
Gravity	g	LT^{-2}
Weir width	b	L
Water height	H	L

There are $n = 4$ variables, and mass does not appear anywhere, so only $r = 2$ primary dimensions (L, T) are involved.

Step 2: Number of dimensionless groups

By the Buckingham Pi theorem, the number of independent dimensionless groups is

$$n - r = 4 - 2 = 2$$

Step 3: Choose repeating variables and form the groups

Choose g and H as repeating variables (together they span both L and T, and neither is the dependent variable Q).

Group 1 (combine with Q):

$$\Pi_1 = \frac{Q}{g^a H^b}$$

Dimensions: $[Q] = L^3T^{-1}$, $[g^a] = L^aT^{-2a}$, $[H^b] = L^b$. Matching powers of L and T:

$$L:\ 3 = a + b \qquad T:\ -1 = -2a \implies a = \tfrac{1}{2}$$

$$b = 3 - \tfrac{1}{2} = \tfrac{5}{2}$$

$$\Pi_1 = \frac{Q}{\sqrt{g}\, H^{5/2}}$$

Group 2 (combine with b): since b is already a length, exactly like H, it forms a dimensionless ratio on its own:

$$\Pi_2 = \frac{b}{H}$$

Step 4: General dimensionless relation

$$\Pi_1 = \phi(\Pi_2) \quad \implies \quad \frac{Q}{\sqrt{g}\, H^{5/2}} = \phi\Big(\frac{b}{H}\Big)$$

$$Q = \sqrt{g}\, H^{5/2}\, \phi\Big(\frac{b}{H}\Big)$$

Step 5: Apply the extra physical information $Q \propto b$

For Q to come out *linearly* proportional to b while b only enters through the function $\phi(b/H)$, that function itself must be linear in its argument:

$$\phi\Big(\frac{b}{H}\Big) = C \cdot \frac{b}{H}$$

where C is a dimensionless constant (not determined by dimensional analysis alone – it must come from experiment or a separate theoretical model). Substituting:

$$Q = \sqrt{g}\, H^{5/2} \cdot C\, \frac{b}{H} = C\, b\, \sqrt{g}\, H^{3/2}$$

Final Answer

$$\boxed{Q = C\, b\, g^{1/2} H^{3/2}}$$

Physical Check

This matches the familiar empirical rectangular-weir formula used in hydraulics, $Q = C_d\, b\sqrt{2g}\, H^{3/2}$ (with all constants absorbed into C) – exactly the structural form predicted by dimensional analysis, with only the discharge coefficient C left to be found experimentally.

Problem 4 — Cone-Plate Viscometer by the Pi Theorem

Given: $M = f(R, \Omega, \mu, \theta)$, where M is the torque needed to rotate the cone, R is the cone radius, Ω is the rotation rate, μ is the fluid viscosity, and θ is the (already dimensionless) cone angle. Rewrite in dimensionless form, then simplify using the fact that $M \propto \mu$.

Step 1: List variables and dimensions

Variable	Symbol	Dimensions (MLT)
Torque	M	ML^2T^{-2}
Radius	R	L
Rotation rate	Ω	T^{-1}
Viscosity	μ	$ML^{-1}T^{-1}$
Cone angle	θ	dimensionless

θ is already dimensionless, so it is automatically its own Pi group and takes no further part in forming the others. Among the remaining 4 dimensional variables (M, R, Ω, μ), $r = 3$ primary dimensions (M, L, T) appear, so:

$n - r = 4 - 3 = 1$ additional Pi group from the dimensional variables, plus $\theta \Rightarrow 2$ groups total

Step 2: Form the dimensionless torque group

Choose R, Ω, μ as repeating variables (they cover L, T, and M independently) and combine with M:

$$\Pi_1 = \frac{M}{\mu\, R^a\, \Omega^b}$$

Dimensions: $[M] = ML^2T^{-2}$, $[\mu] = ML^{-1}T^{-1}$, $[R^a] = L^a$, $[\Omega^b] = T^{-b}$.

$$\frac{ML^2T^{-2}}{(ML^{-1}T^{-1})(L^a)(T^{-b})} = M^{1-1}L^{2-(-1+a)}T^{-2-(-1-b)} = L^{3-a}\,T^{-1+b}$$

Setting exponents to zero for dimensionlessness:

$$L:\ 3 - a = 0 \Rightarrow a = 3 \qquad\qquad T:\ -1 + b = 0 \Rightarrow b = 1$$

$$\Pi_1 = \frac{M}{\mu R^3 \Omega}$$

Step 3: General dimensionless relation

$$\boxed{\frac{M}{\mu\, \Omega\, R^3} = f(\theta)}$$

Step 4: Simplification using $M \propto \mu$

Look carefully at the four original dimensional variables: M, R, Ω, μ. Among these, μ **is the only one that carries the dimension of mass**. Since M itself carries mass to the first power, *any* dimensionally homogeneous relation between these variables is automatically forced to have M proportional to μ raised to exactly the first power – there is no other way to balance the mass dimension. In other words, the Pi-theorem result already derived,

$$M = \mu\, \Omega\, R^3\, f(\theta),$$

is *already* linear in μ by construction.

Final Answer

Being told separately that "M is proportional to μ" therefore provides **no new information** and causes **no further simplification** of the relation – it merely confirms what dimensional analysis had already guaranteed. The most general form remains

$$\boxed{M = \mu\,\Omega\,R^3\,f(\theta)}$$

with the shape of $f(\theta)$ left undetermined by dimensional analysis alone (it must come from experiment or a full analytical solution of the flow inside the cone-plate gap).

Physical Check

(For reference/interest, beyond pure dimensional analysis: an exact thin-gap Newtonian-flow solution for the cone-and-plate geometry gives $M = \dfrac{2\pi}{3}\dfrac{\mu\Omega R^3}{\theta}$ for small θ, which indeed has the predicted form $M = \mu\Omega R^3 f(\theta)$ with $f(\theta) = 2\pi/(3\theta)$.)

Problem 5 — Feasibility of a 1/20-Scale Wind-Tunnel Test

Given: Prototype (full-scale airplane) velocity $V_p = 300$ mi/h. A model at length scale $L_m/L_p = 1/20$ is to be tested in a wind tunnel at the *same* pressure and temperature as the full-scale flight (i.e., using air with the same density ρ and viscosity μ in both cases). Is it feasible to determine the prototype drag coefficient this way?

Concept Refresher

For dynamic similarity of two geometrically similar flows (so that the measured drag coefficient $C_D = D/(\frac{1}{2}\rho V^2 L^2)$ is the same for model and prototype), the governing dimensionless group – the Reynolds number – must match:

$$Re_m = Re_p \quad \Longrightarrow \quad \frac{\rho_m V_m L_m}{\mu_m} = \frac{\rho_p V_p L_p}{\mu_p}$$

Step 1: Simplify using "same pressure and temperature"

Same pressure and temperature for the same fluid (air) means $\rho_m = \rho_p$ and $\mu_m = \mu_p$ (density and viscosity are functions only of pressure and temperature for a given gas). These cancel from the Reynolds-number matching condition:

$$V_m L_m = V_p L_p$$

Step 2: Solve for the required model velocity

$$V_m = V_p \frac{L_p}{L_m} = V_p \times 20$$

$$V_m = 300 \times 20 = 6000 \text{ mi/h}$$

Step 3: Assess feasibility

A required test-section velocity of 6000 mi/h is far beyond what any conventional (or even most specialized) wind tunnel can produce. Moreover, this speed corresponds to a very high Mach number:

$$a \approx 761 \text{ mi/h (speed of sound at standard conditions)}$$

$$Ma_m = \frac{6000}{761} \approx 7.9 \qquad \text{(strongly hypersonic!)}$$

while the actual prototype flies at

$$Ma_p = \frac{300}{761} \approx 0.39 \qquad \text{(subsonic, compressibility negligible)}$$

Even if such a velocity *could* somehow be reached, the model flow would then be dominated by compressibility, shock waves, and aerodynamic heating effects that are completely absent in the real (subsonic) prototype flow – so *matching Reynolds number this way would badly violate Mach-number similarity*, and the measured model drag coefficient would **not** correspond to the prototype's.

Final Answer

No, it is not feasible. Matching Reynolds number for a 1/20-scale model at the same pressure and temperature would require an unattainable test velocity of about 6000 mi/h (Mach ≈ 7.9), which is both practically impossible for standard wind tunnels and physically inappropriate (it would introduce severe compressibility effects not present in the actual

subsonic prototype flow, at Mach ≈ 0.39). In practice, engineers instead achieve the required Reynolds number at a feasible, low-subsonic model velocity by using a **pressurized wind tunnel** (raising ρ) and/or a denser test fluid, rather than raising velocity alone.